More on matrices and linear equations.

Let $A \in M_{m\times n}(K)$, and consider the system of linear eqs.

$A\cdot x = 0$. Denote by $Sol(A,0) := \{x \in K^n \mid Ax = 0\} = Ker(T_A)$,

the <u>space</u> of solutions of $Ax=0$. Note that

$\dim Sol(A,0) = n - rank(A)$ (b.c. $\dim Ker(T_A) = n - \dim Im(T_A)$).

Let A' be a row-reduced echelon mat. which is row-equiv. to A (e.g. A' is obtained from A by Gauss elimination).

$\#\,pivots(A') = rank(A)$, $n - rank(A) = \#$ free varibles.

$$\begin{pmatrix} 0\cdots 0 & 1 * \cdots * & & & \\ & & 1 * \cdots * & & \\ & & & 1 * \cdots * & \\ & & & & 1 * \cdots * \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

A'

Green: leading 0's in 1'st row and $*$'s correspond to free variables.

Red: pivots.

$$\boxed{Sol(A,0) = Sol(A',0)}$$

Denote by $x_{i_1}, \dots, x_{i_\ell}$ the free variables, where $1 \le i_1 < \dots < i_\ell \le n$, $\ell = n - rank(A)$ and by $x_{j_1}, \dots, x_{j_r}$, $1 \le j_1 < \dots < j_r \le n$, $r = rank(A)$ the pivots.

For each choice of values for the free variables $x_{i_1}, \dots, x_{i_\ell} \in K$, the pivot variables $x_{j_1}, \dots, x_{j_r}$ are uniquely determined. So, we obtain a map

$$\Phi : K^\ell \longrightarrow Sol(A,0), \quad \ell = n - rank(A)$$

$\Phi(\lambda_1, \dots, \lambda_\ell)$ = the unique solut. of $Ax=0$ (= solut. of $A'x=0$)

with $x_{i_1} = \lambda_1, \dots, x_{i_k} = \lambda_k, \dots, x_{i_\ell} = \lambda_\ell$.

Prop. (17.d.1) The map Φ is linear. Moreover it is an iso.

Proof. Linearity of Φ. The k'th non-zero row of A' gives the following equation:

x_{j_k} + linear combin. of the free vars. that come after j_k $= 0$.

$$\underset{\text{(pivot)}}{x_{j_k}} = \sum_{\{q \mid 1 \le q \le \ell,\ i_q > j_k\}} -a'_{k,i_q} \cdot x_{i_q} \leftarrow \text{(free variables)}$$

$\Rightarrow x_{j_1}, \dots, x_{j_r}$ depend linearly on the free variables $x_{i_1}, \dots, x_{i_\ell}$. This proves the linearity of Φ.

To prove that Φ is an iso., it's enough to show that $\mathrm{Ker}(\Phi) = 0$, b.c. $\Phi : K^\ell \longrightarrow \mathrm{Sol}(A,0) = \ell$-dim space.

But this clearly holds: if $\lambda = (\lambda_1, \dots, \lambda_\ell) \in \mathrm{Ker}(\Phi)$, i.e. if $\Phi(\lambda_1, \dots, \lambda_\ell) = 0$, then by def. of Φ we have $\lambda_1 = \dots = \lambda_\ell = 0$ b.c. each λ_k is an entry in $\Phi(\lambda_1, \dots, \lambda_\ell)$. ∎

The inverse map Φ^{-1} of Φ: let $x \in \mathrm{Sol}(A,0)$,

$\Phi^{-1}(x) = \begin{pmatrix} x_{i_1} \\ \vdots \\ x_{i_\ell} \end{pmatrix}$, where $x_{i_1}, \dots, x_{i_\ell}$ are the entries in x corresp. to the free variables.

Cor. (17.d.2) Let $b \in K^m$, $A \in M_{m\times n}(K)$. Then:

1) $Sol(A,b) := \{x \in K^n \mid A\cdot x = b\} \neq \emptyset$ iff $b \in ColS(A)$.

2) If $Sol(A,b) \neq \emptyset$ and $y \in Sol(A,b)$ then

$$Sol(A,b) = y + Sol(A,0) := \{y + x \mid x \in Sol(A,0)\}.$$

Proof. Exc.

Hints: 1) If $A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$, $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, then

$$A\cdot x = x_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + \dots + x_n \cdot \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}.$$

2) Let $y \in Sol(A,b)$.

If $x \in Sol(A,0)$ then $A\cdot x = 0$, $A\cdot y = b$.

$\Rightarrow A(y+x) = A\cdot y + A\cdot x = b + 0 = b$.

Conversely, if $z \in Sol(A,b)$, i.e. $A\cdot z = b$, then define $x := z - y$. We have $z = y + x$ and $A\cdot x = A\cdot z - A\cdot y = b - b = 0$. ∎

Prop. (17.d.3) Let $A \in M_{m\times n}(K)$, $b \in K^m$. The following statements are equivalent:

1) $\text{rank}(A|b) = \text{rank}(A)$.

2) The system of eqs. $Ax = b$ has a solution.

Proof. Write $A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$. We have:

$1 \Leftrightarrow \text{col-rank}(A|b) = \text{col-rank}(A) \Leftrightarrow Sp(v_1,\dots,v_n,b) = Sp(v_1,\dots,v_n)$

$\Leftrightarrow b \in Sp(v_1,\dots,v_n) \Leftrightarrow b \in ColS(A) \Leftrightarrow Sol(A,b) \neq \emptyset$. ∎

Prop. (17.d.4) Let $A \in M_{m\times n}(K)$, $b \in K^m$. The following statements are equivalent:

1) $\operatorname{rank}(A) = \operatorname{rank}(A|b) = n$.

2) The system $Ax=b$ has a unique solution.

Proof. Exc.

Hints: Write $A = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}$. $\operatorname{rank}(A) = n \iff \dim Sp(v_1,\dots,v_n) = n$ $\iff v_1,\dots,v_n$ form a basis for $Col\,S(A)$. ∎